7.Write a program to implement Logistic Regression (LR) algorithm in python

CODING:

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

X = np.array([

    [1, 50],

    [2, 55],

    [2, 60],

    [3, 65],

    [3, 70],

    [4, 72],

    [4, 75],

    [5, 80],

    [5, 85],

    [6, 88],

    [6, 90],

    [7, 92],

    [8, 95],

    [9, 96],

    [10, 98]

])

y = np.array([

    0, 0, 0, 0, 0,

    1, 1, 1, 1, 1,

    1, 1, 1, 1, 1
```

```
])
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
model = LogisticRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("Logistic Regression Results")
print("---------------------------")
print("Actual Values   :", y_test)
print("Predicted Values:", y_pred)
print("Accuracy        :", accuracy)
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
new_student = np.array([[5, 82]])
prediction = model.predict(new_student)
if prediction[0] == 1:
    print("\nNew Student Prediction: PASS")
else:
    print("\nNew Student Prediction: FAIL")
```

OUTPUT:

```
Logistic Regression Results
---------------------------
Actual Values   : [1 1 0 1 1]
Predicted Values: [1 1 0 1 0]
Accuracy        : 0.8

Confusion Matrix:
[[1 0]
 [1 3]]

Classification Report:
              precision    recall  f1-score   support

           0       0.50      1.00      0.67         1
           1       1.00      0.75      0.86         4

    accuracy                           0.80         5
   macro avg       0.75      0.88      0.76         5
weighted avg       0.90      0.80      0.82         5
```